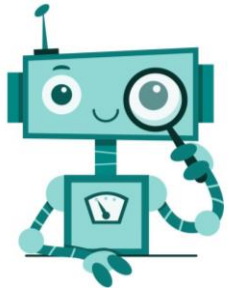
Artificial Intelligence (AI)



AGENDA

- Critical Thinking
- Neural Network
- Perceptron
- MLP
- Types of Neural Network

What is this?

You see this:



But the camera sees this:

194	210	201	212	199	213	215	195	178	158	182	209
180	189	190	221	209	205	191	167	147	115	129	163
114	126	140	188	176	165	152	140	170	106	78	88
87	103	115	154	143	142	149	153	173	101	57	57
102	112	106	131	122	138	152	147	128	84	58	66
94	95	79	104	105	124	129	113	107	87	69	67
68	71	69	98	89	92	98	95	89	88	76	67
41	56	68	99	63	45	60	82	58	76	75	65
20	43	69	75	56	41	51	73	55	70	63	44
50	50	57	69	75	75	73	74	53	68	59	37
72	59	53	66	84	92	84	74	57	72	63	42
67	61	58	65	75	78	76	73	59	75	69	50

Computer Vision: Car detection








Cars








Not a car

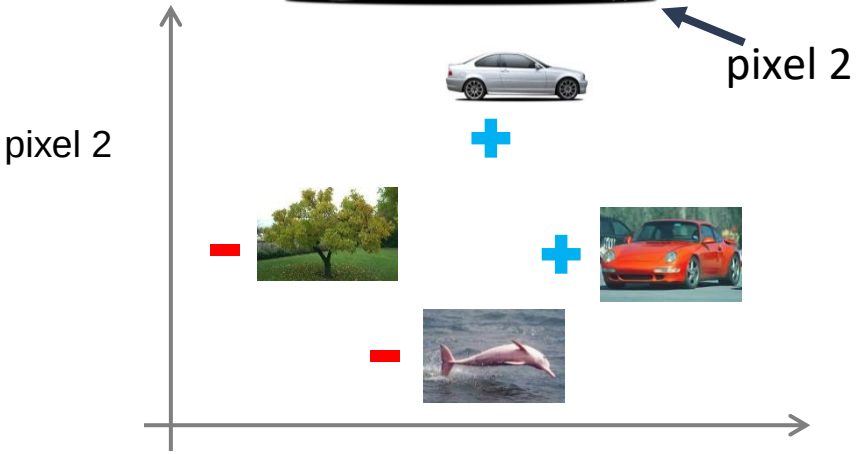
Testing:
What is this?



Learning Algorithm



Learning Algorithm



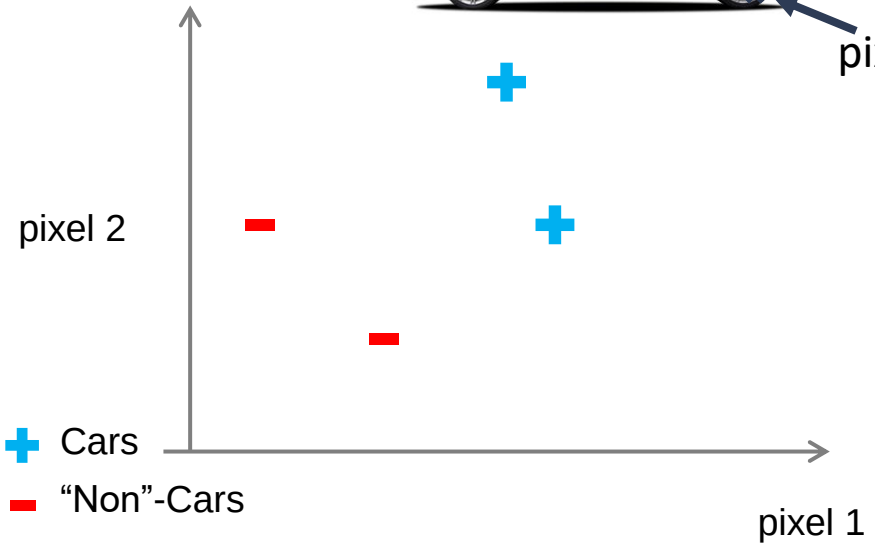
+ Cars

– “Non”-Cars

Learning Algorithm

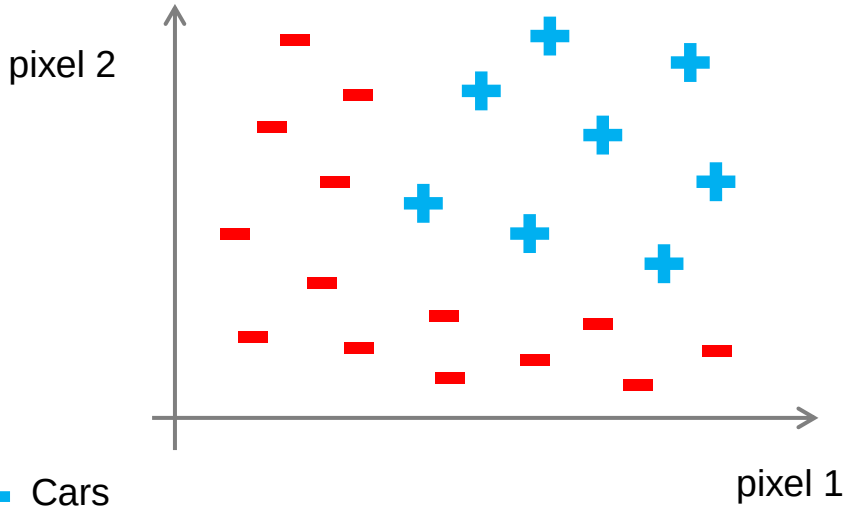


Learning Algorithm





Learning Algorithm



50 x 50 pixel images→ 2500 pixels
n= 2500 (7500 if RGB)

$$x = \begin{bmatrix} \text{pixel 1 intensity} \\ \text{pixel 2 intensity} \\ \vdots \\ \text{pixel 2500 intensity} \end{bmatrix}$$

Quadratic features ($x_i \times x_j$): ≈3 million features

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Neural Networks: Representation

Neurons and the brain

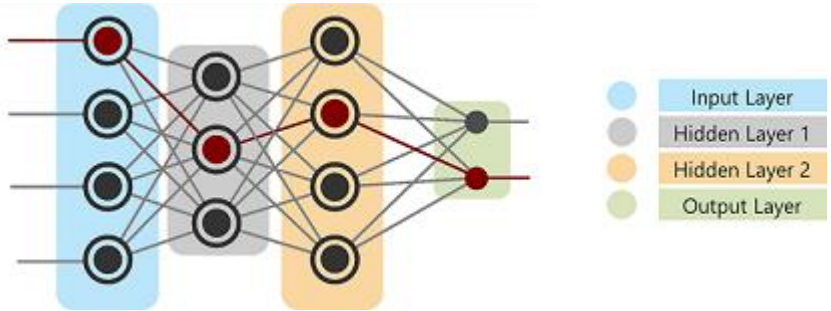
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Neural Network Origin

Origins: Algorithms that try to mimic the brain.
Was very widely used in 80s and early 90s; popularity diminished in late 90s.
Recent resurgence: State-of-the-art technique for many applications

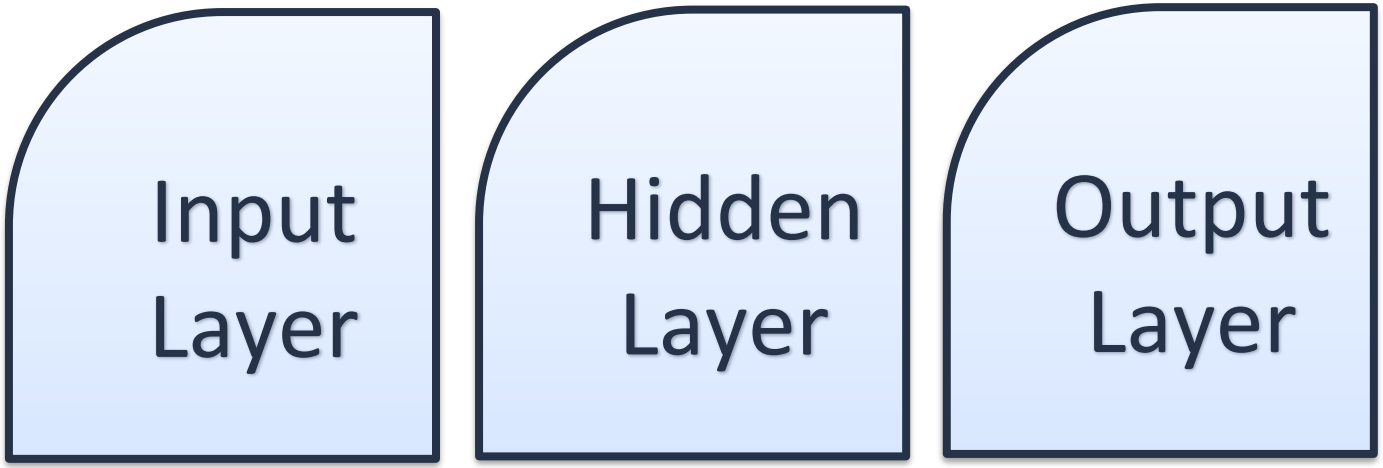
Definition Of A Neural Network

- Modeled in accordance with the human brain, a Neural Network was built to mimic the functionality of a human brain. The human brain is a neural network made up of multiple neurons, similarly, an Artificial Neural Network (ANN) is made up of multiple perceptron's (explained later).



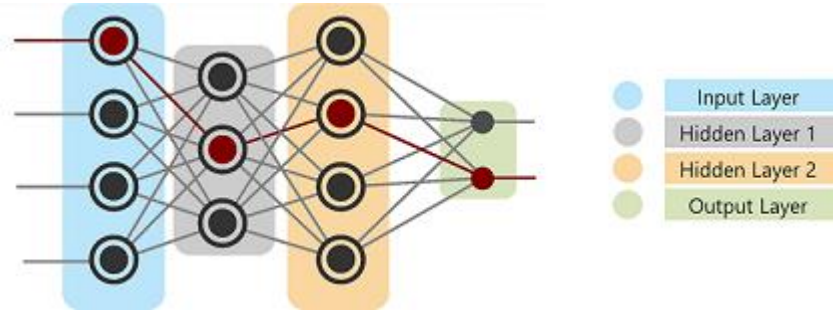
Layers in Neural Network

A neural network consists of three important layers:



hidden layer

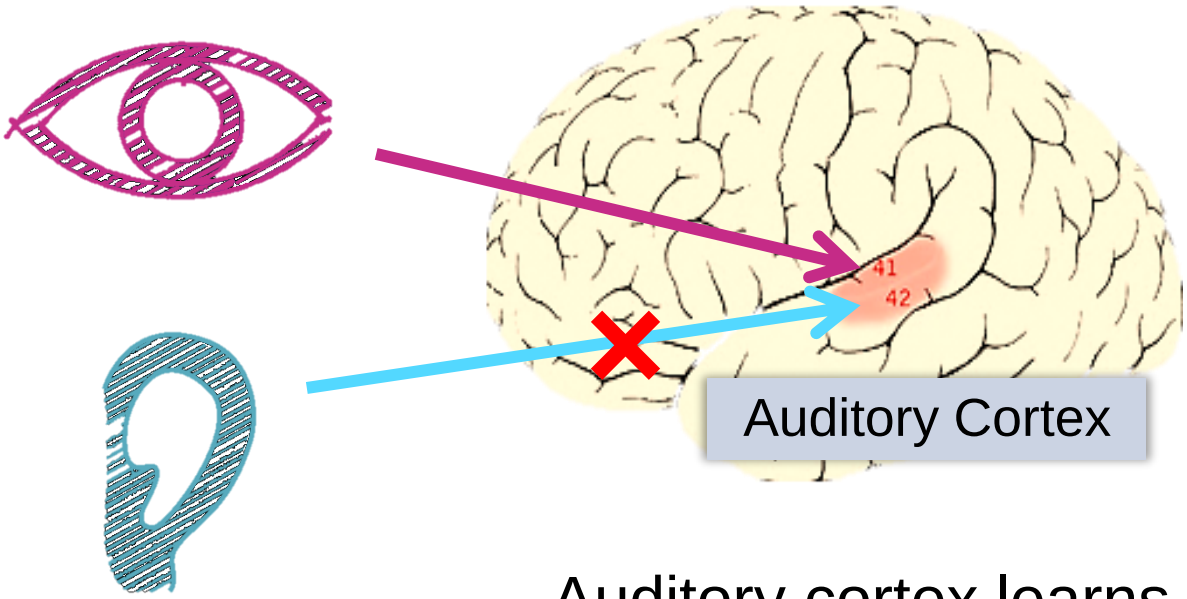
- A hidden layer in an artificial neural network is a layer in between input layers and output layers, where artificial neurons take in a set of weighted inputs and produce an output through an activation function.



Activation Function

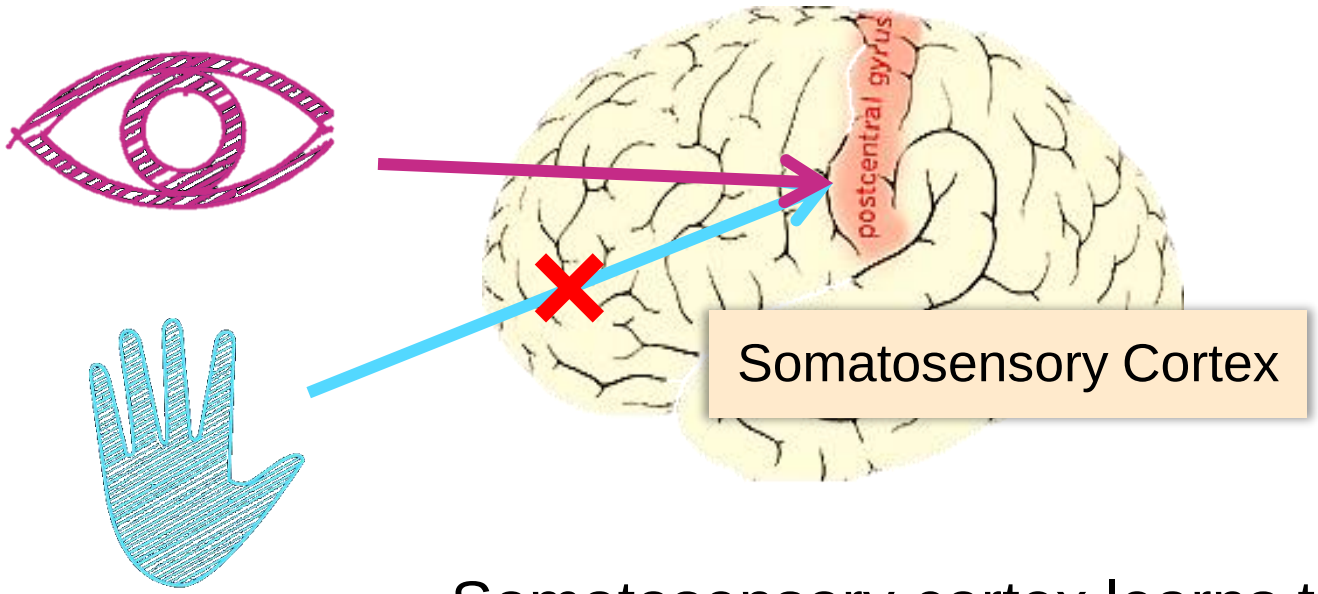
- An activation function in a neural network defines how the weighted sum of the input is transformed into an output from a node or nodes in a layer of the network

The “one learning algorithm” hypothesis



Auditory cortex learns to see

The “one learning algorithm” hypothesis



Somatosensory cortex learns to see

Sensor representations in the brain

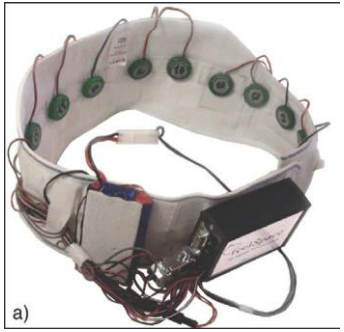




Seeing with your tongue

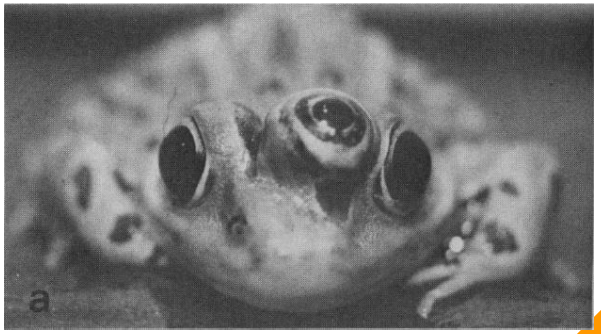


Human echolocation (sonar)



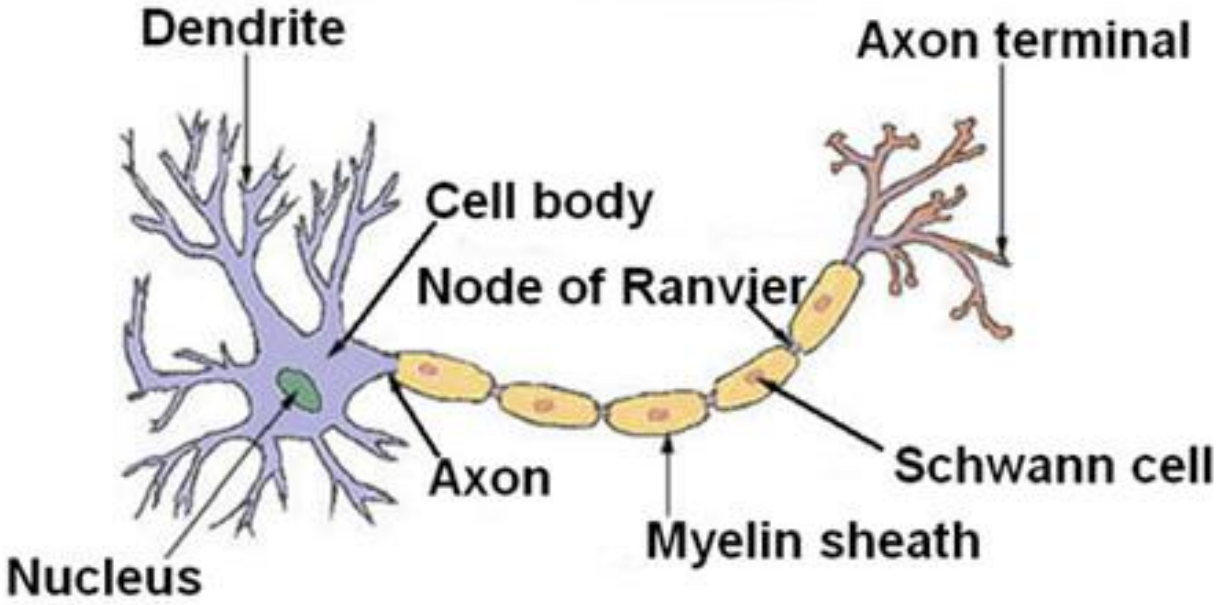


Haptic belt: Direction sense



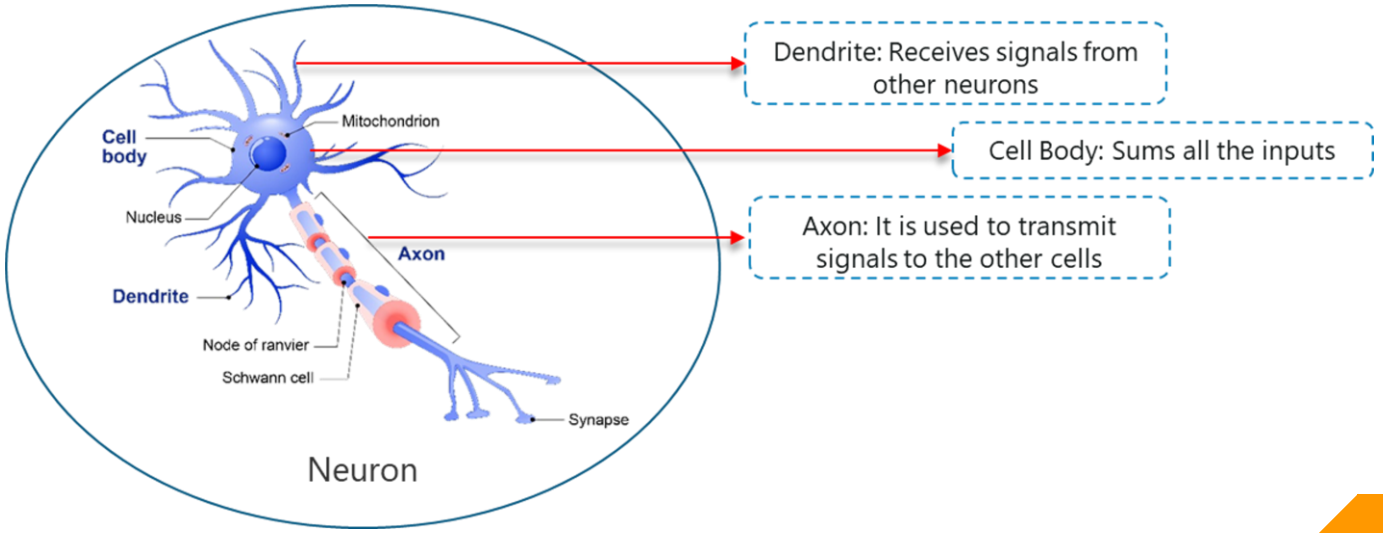
Implanting a 3rd eye

Neuron in the brain



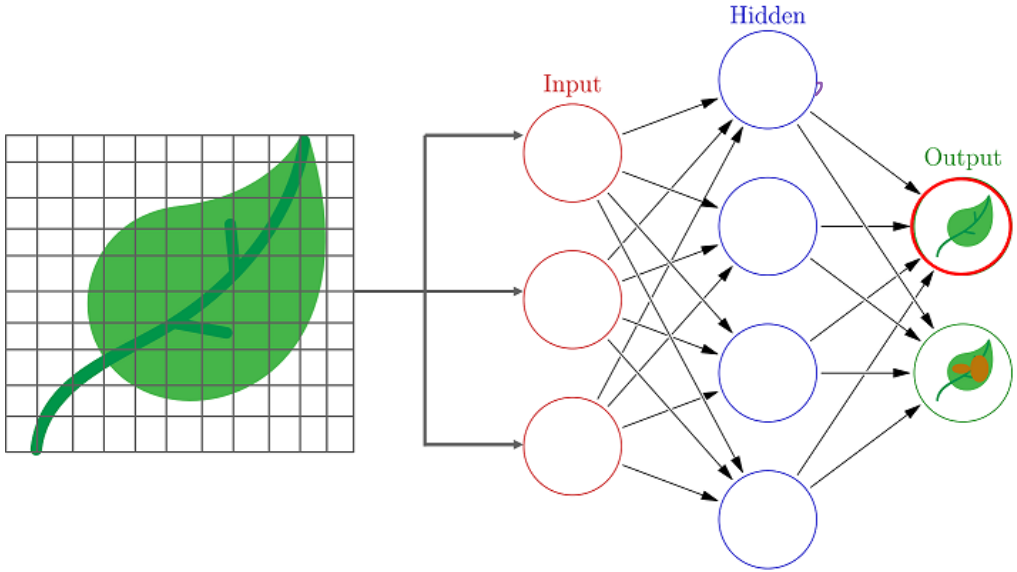
Motivation behind neural network

The building block of a neural network is the neuron. An artificial neuron works much the same way the biological one days.

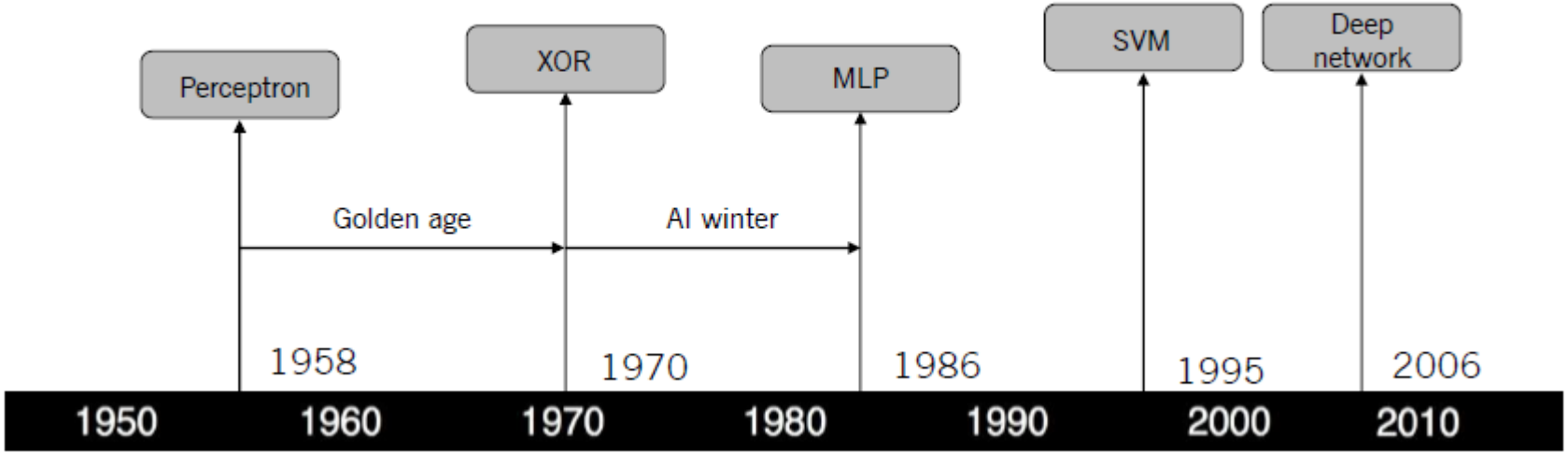


Neural Networks Explained With An Example

- Class A: Containing images of non-diseased leaves
- Class B: Containing images of diseased leaves

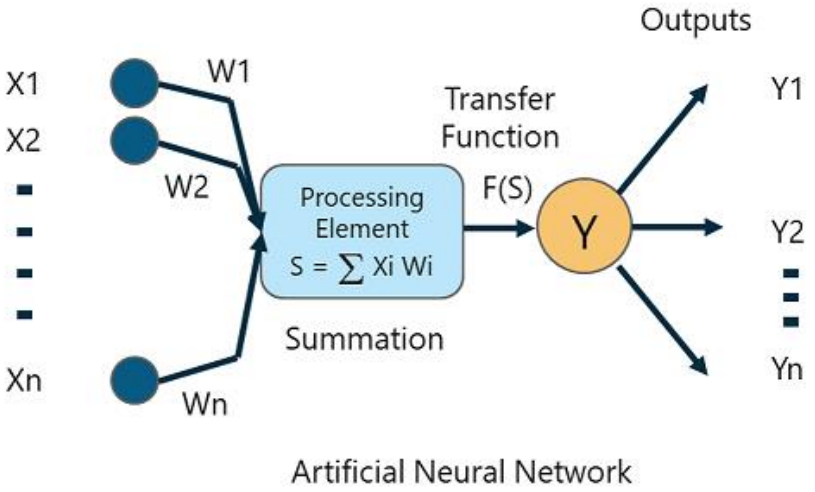


Development History of Neural Networks



What Is A Perceptron?

- A Perceptron is a single layer neural network that is used to classify linear data. It has 4 important components:
  - Inputs
  - Weights and Bias
  - Summation Function
  - Activation or transformation Function



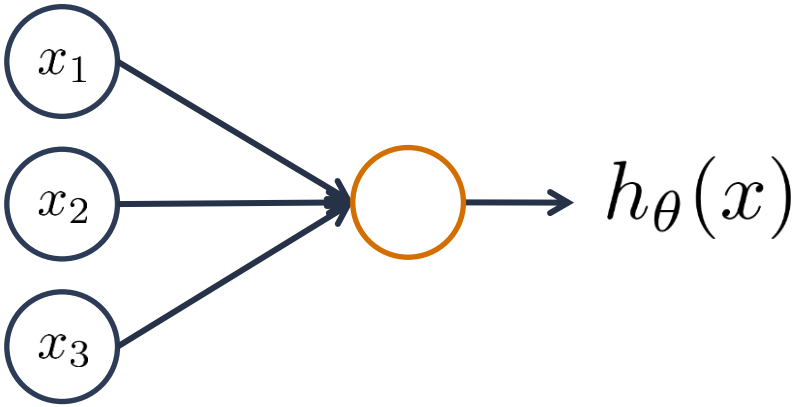
Artificial Neural Network

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Limitations of Single-Layer Perceptron:

- Single-Layer Perceptron cannot classify non-linearly separable data points.
- Complex problems, that involve a lot of parameters cannot be solved by Single-Layer Perceptron.

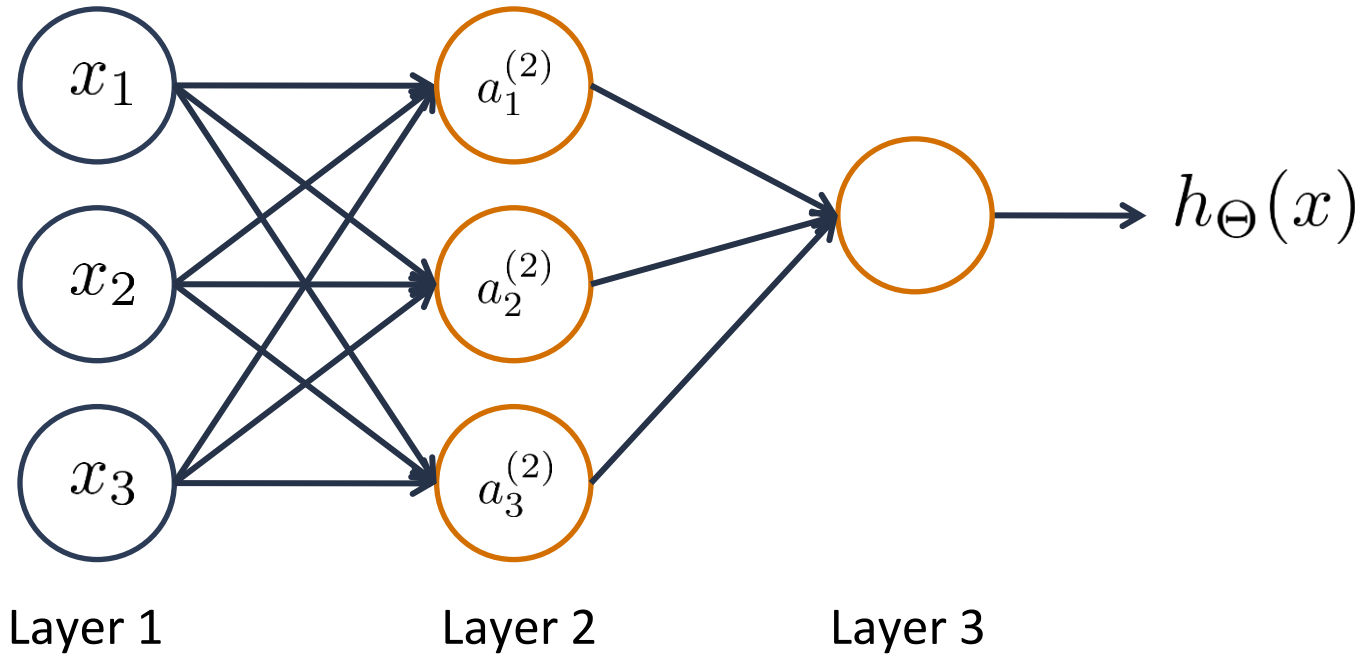
Neuron model: Logistic unit



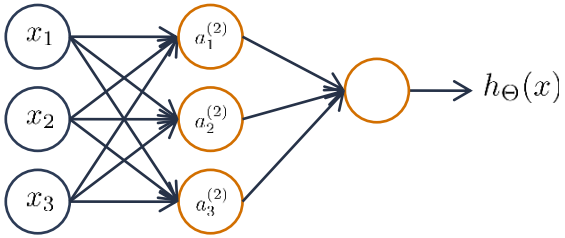
$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$$

Sigmoid (logistic) activation function.

Neural Network



Neural Network



$a_i^{(j)}$ = "activation" of unit i in layer j

$\Theta^{(j)}$ = matrix of weights controlling function mapping from layer j to layer $j+1$

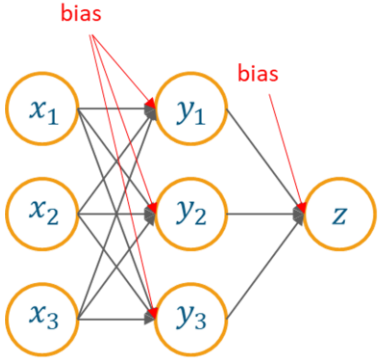
$$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$$
$$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$$
$$a_3^{(2)} = g(\Theta_{30}^{(1)}x_0 + \Theta_{31}^{(1)}x_1 + \Theta_{32}^{(1)}x_2 + \Theta_{33}^{(1)}x_3)$$
$$h_\Theta(x) = a_1^{(3)} = g(\Theta_{10}^{(2)}a_0^{(2)} + \Theta_{11}^{(2)}a_1^{(2)} + \Theta_{12}^{(2)}a_2^{(2)} + \Theta_{13}^{(2)}a_3^{(2)})$$

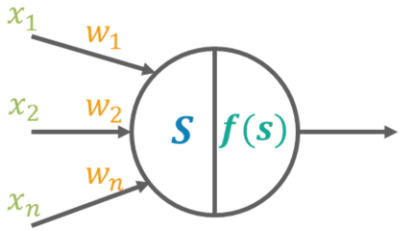
If network has s_j units in layer j, s_{j+1} units in layer $j+1$, then $\Theta^{(j)}$ will be of dimension $s_{j+1} \times (s_j + 1)$.

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What Is A Multilayer Perceptron?

- A Multilayer perceptron is a classifier that contains one or more hidden layers and it is based on the Feedforward artificial neural network. In the Feedforward networks, each neural network layer is fully connected to the following layer.





Summation:

$$S = \sum_{i=1}^{n} w_i * x_1$$

Transformation:

$$f(x) = \frac{1}{1+e^{-\beta x}}$$

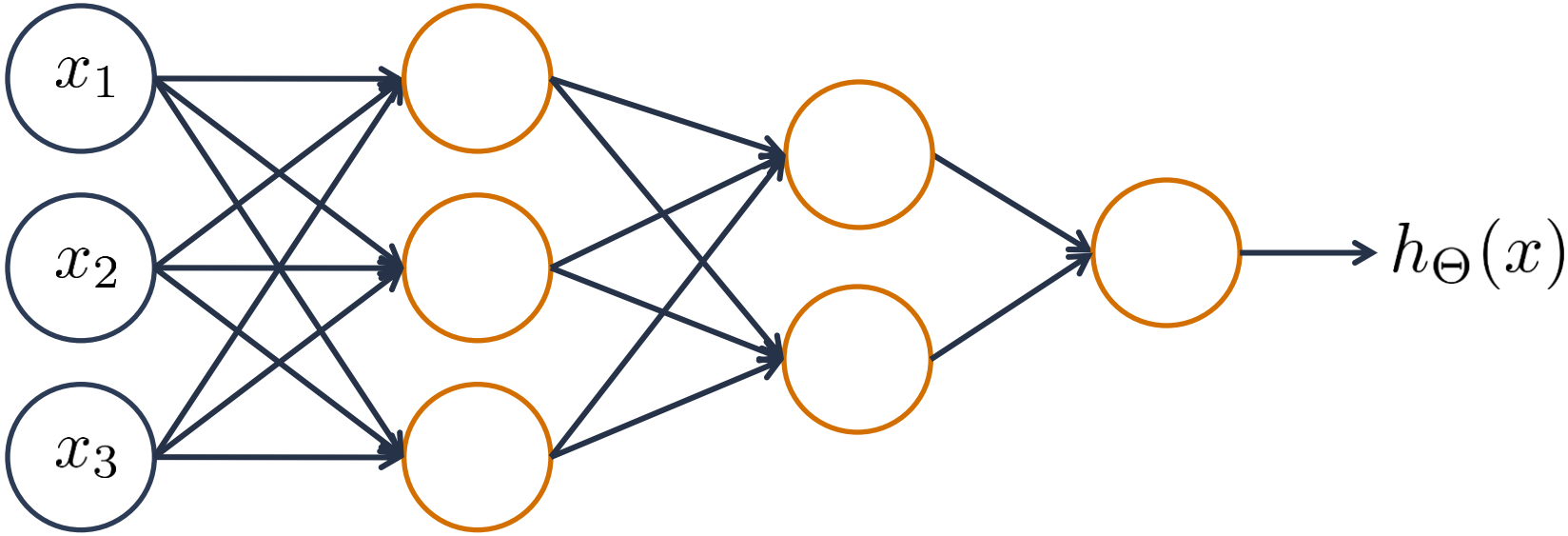
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Types of Neural Network

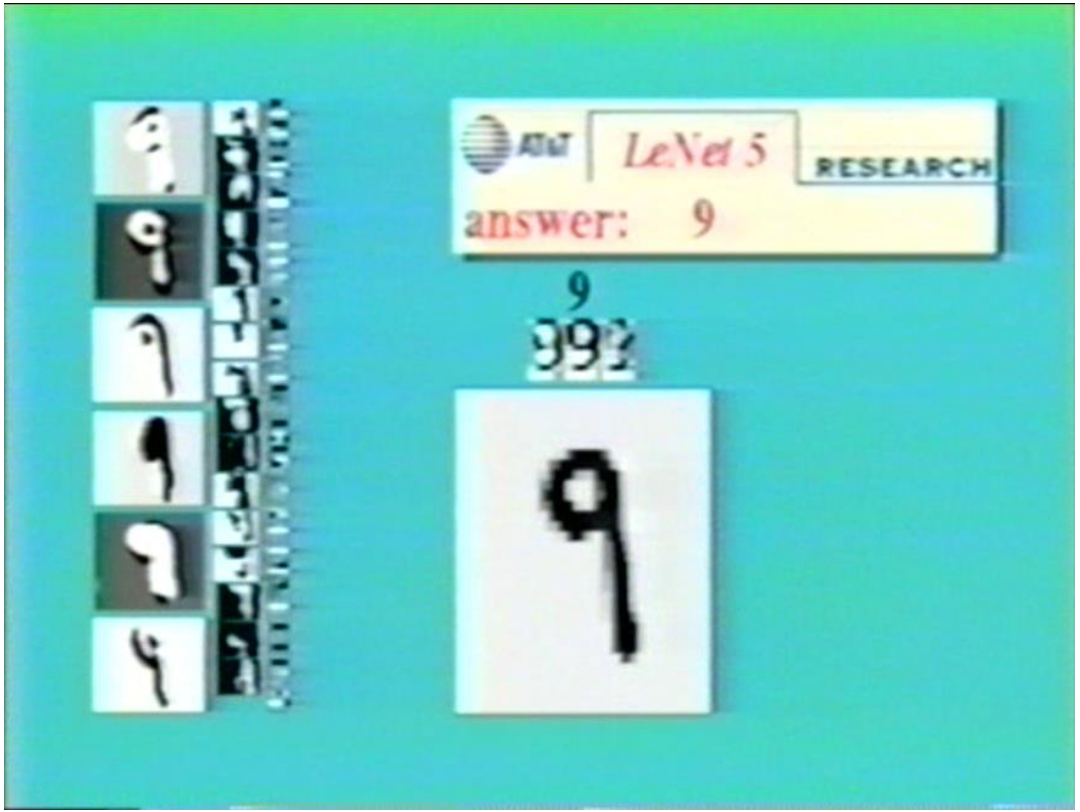
- ANN (Artificial neural network)
- CNN (Convolutional neural network)
- RNN (Recurrent neural network)

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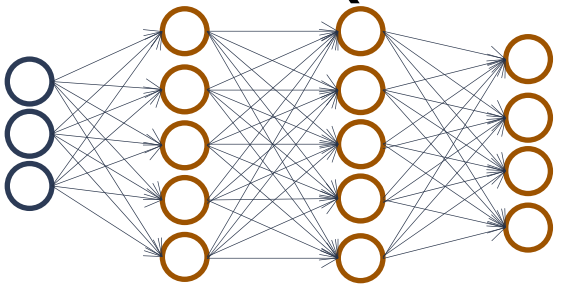
Neural Network intuition



Handwritten digit classification



Neural Network (Classification)



$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(m)}, y^{(m)})\}$

$L =$ total no. of layers in network

$s_l =$ no. of units (not counting bias unit) in layer l

Binary classification

$y = 0 \text{ or } 1$

1 output unit

Multi-class classification (K classes)

$y \in \mathbb{R}^K$

E.g. $\begin{bmatrix}1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\1\end{bmatrix}$

pedestrian car motorcycle truck

K output units

Cost function

Logistic regression:

$$J(\theta) = -\frac{1}{m}\left[\sum_{i=1}^{m} y^{(i)} \log h_\theta(x^{(i)}) + (1 - y^{(i)}) \log(1 - h_\theta(x^{(i)}))\right] + \frac{\lambda}{2m}\sum_{j=1}^{n} \theta_j^2$$

Neural network:

$$h_\Theta(x) \in \mathbb{R}^K \quad (h_\Theta(x))_i = i^{th} \text{ output}$$

$$J(\Theta) = -\frac{1}{m}\left[\sum_{i=1}^{m}\sum_{k=1}^{K} y_k^{(i)} \log((h_\Theta(x^{(i)}))_k + (1 - y_k^{(i)}) \log(1 - (h_\Theta(x^{(i)}))_k)\right]$$

$$+\frac{\lambda}{2m}\sum_{l=1}^{L-1}\sum_{i=1}^{s_l}\sum_{j=1}^{s_{l+1}}(\Theta_{ji}^{(l)})^2$$

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